

Sequential Data and RNN Architecture

Deep Learning

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Contents of This Video

In this video, we will cover:

- What makes sequence data special
- Why feedforward networks fail for sequences
- The recurrent neural network architecture
- Unrolling an RNN through time
- Types of RNN tasks

Sequential Data Examples

Text: “I love machine learning”

- Order matters: “learning machine love I” is gibberish
- Each word depends on previous words

Time series: Stock prices, sensor readings, weather

- Current value depends on history
- Patterns unfold over time

Speech: Audio waveforms

- Meaning emerges from sequence of sounds
- Context from previous sounds

Why Feedforward Networks Fail

Problem 1: Fixed input size

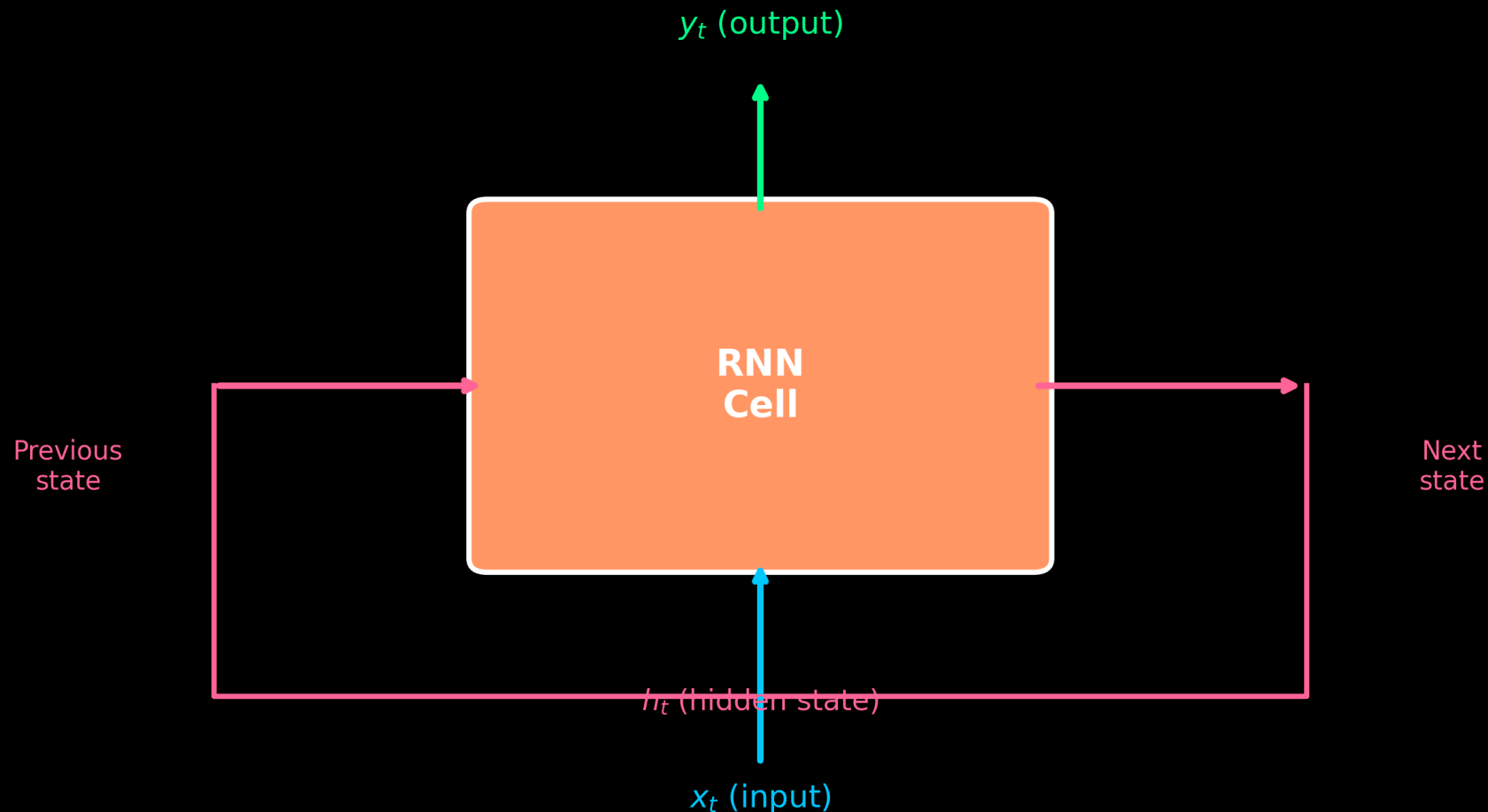
- FC networks need fixed-size input
- But sequences have variable length
- How do you handle a 5-word vs. 50-word sentence?

Problem 2: No memory

- Each input processed independently
- Can't remember previous elements
- Context is lost

The RNN Idea: Persistent State

RNN: Hidden State Creates Memory



Key idea: Hidden state carries information forward in time

RNN Cell Computation

At each timestep t :

$$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t + b_h)$$

$$y_t = W_{hy} h_t + b_y$$

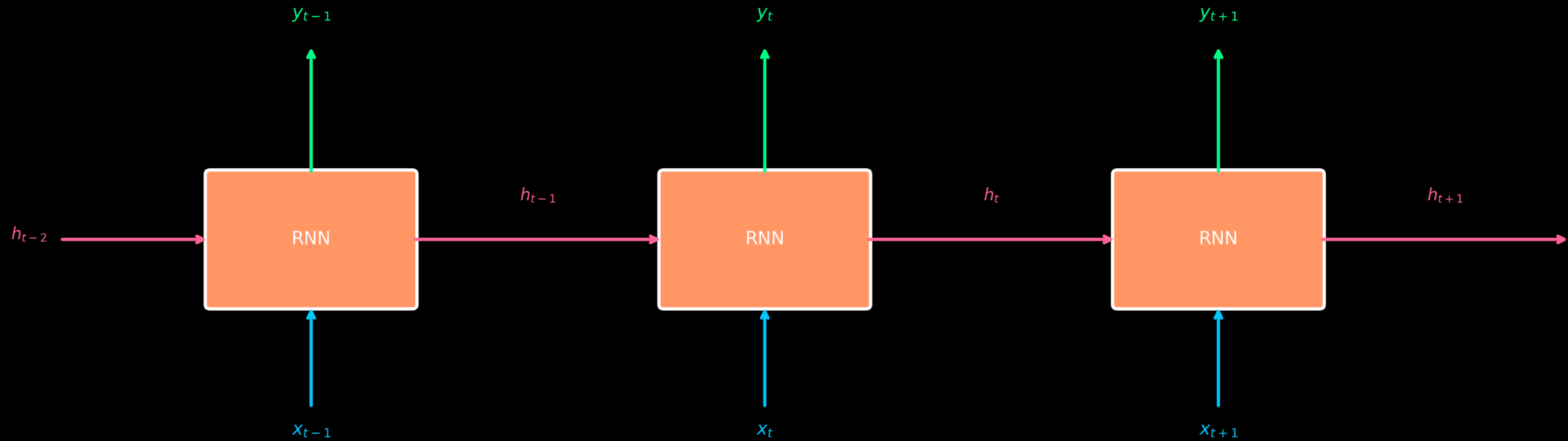
Where:

- x_t : input at time t
- h_t : hidden state at time t
- y_t : output at time t
- W_{hh} , W_{xh} , W_{hy} : weight matrices (shared across time!)
- $\tanh$: activation function

Unrolling Through Time

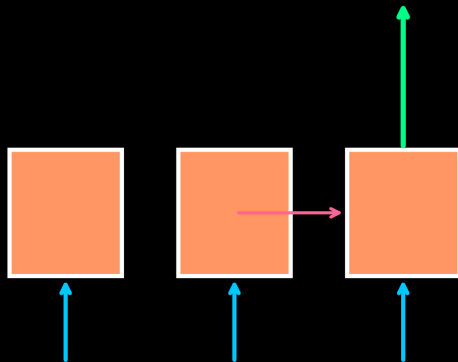
RNN Unrolled Through Time

Same weights at every timestep!

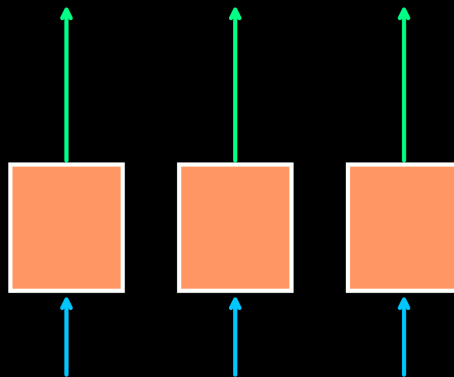


Types of RNN Tasks

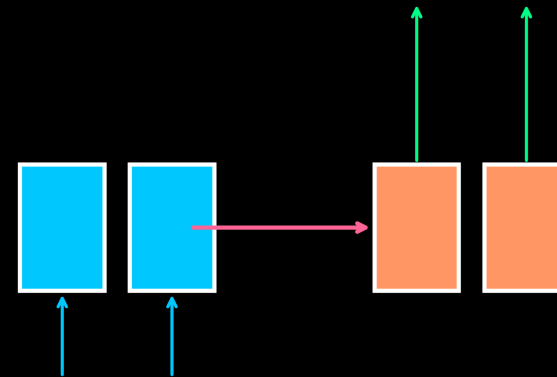
Many-to-One
(Sentiment)



Many-to-Many
(POS Tagging)



Encoder-Decoder
(Translation)



What We've Covered

In this video, we've learned:

- Sequential data: order matters (text, time series, speech)
- Feedforward networks lack memory for sequences
- RNNs use hidden state to carry information forward
- $h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t)$
- Same weights shared across all timesteps
- Different RNN architectures for different tasks